Light Echo Modeling of the Active Galactic Nucleus NGC 3783 Using TORMAC - WORKING DRAFT

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ABSTRACT
When supermassive black holes, found in the center of most galaxies, rapidly consume material, they are known as active galactic nuclei (AGN). This study employs computational modeling to explore the dust structure commonly found around AGN, focusing on one such AGN, NGC 3783. Utilizing the TORMAC code developed by Dr. Triana Almeyda, we generate simulated infrared (IR) light curves based on a 3D ensemble of clouds within a simulated dust structure, using observed optical light curves of NGC 3783 as the input continuum source. We use a Markov Chain Monte Carlo (MCMC) code, written by collaborator Dr. Jack Gallimore, to determine the best-fit parameters of the dust distribution by comparing the simulated IR light curves to the observed IR light curves for NGC 3783. Here we present preliminary results from these simulations and demonstrate their application towards constraining the structure and geometry of NGC 3783. [instead of what was here, add some discussion / key findings]

Key words: radiative transfer – galaxies: active

1 INTRODUCTION

At the center of most galaxies resides a super massive black hole (SMBH) with masses typically exceeding $10^6 M_\odot$. When these SMBHs rapidly accrete surrounding gas and dust, they are classified as active galactic nuclei (AGN). Close to the center of these AGN, fueled primarily by the gravitational potential energy of the infalling material, a relatively small disk (the accretion disk) of super heated plasma emits primarily in the X-ray, ultraviolet (UV), and optical portions of the electromagnetic (EM) spectrum. These AGN have luminosities typically between 10^{40} ergs•s-1 and 10^{47} erg•s-1 and can, in some cases, exceed the luminosity of their host galaxy.

Surrounding this accretion disk is a much larger structure of circumnuclear dust which reprocesses the optical continuum radiation and reemits it in the infrared (IR). AGN unification models imply that the observed lack of broad emission lines in the spectra of some AGN is due to the presence of this dust structure [Zach: cite], which obscures the broad line region (BLR) from our line of sight (LOS) in these AGN. Therefore, because the presence of this dust structure is a key component of the AGN unification model, it is important to understand the structural (e.g. the distribution and composition of dust) and geometric (e.g. the size and shape of the dust structure) features of this dust.

Directly resolving the inner radius of the circumnuclear dust is challenging except for the closest AGN, due to the sub-milliarcsecond spatial resolution required. As a result, other methods have been developed to study the circumnuclear dust's structure and geometry. Using IR interferometry, the inner radius and approximate spatial extent of the hot dust has been measured for several nearby AGN **[Cited in Almeyda et al. 2017: Swain et al. 2003; Kishimoto et al. 2009; Pott et al. 2010; Kishimoto et al. 2011; Weigelt et al. 2012], [Cited in Almeyda et al. 2020: e.g., Kishimoto et al. 2011; Burtscher et al. 2013]**. However, this method does not extend well to AGN of higher redshift.

Observational reverberation mapping has been extensively used to constrain the characteristic size of the dust across many AGN [Zach: cite here]. By measuring the average time delay between optical variability and the corresponding IR response using cross correlation analysis, one can obtain a measurement of the characteristic radius of the circumnuclear dust. Observational reverberation mapping avoids the need for high resolution imaging, making it applicable to AGN at higher redshifts. However, this method only provides a single characteristic radius for the dust and does not provide any information about the structure or geometry of the dust.

In reality, the IR dust reprocessing of the optical continuum emission occurs over a range of times, a direct result of dust being at different distances from the central source and at different viewing angles to the observer. This causes the IR response to not be completely correlated with the optical continuum emission which drives it. This means that the IR response light curve also encodes information about the structure and geometry of the dust. By modeling the IR response light curve, we can attempt to extract this additional information about the dust structure and geometry.

The forward modeling code TORMAC (TOrus Reverberation MApping Code) has previously been employed to simulate the multiwavelength response from a circumnuclear dust distribution to a delta function "pulse" from the optical continuum, emulating that dust configuration's "transfer function" (Almeyda et al. 2017, 2020) which

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describes how the IR emission is smeared in time due to a driving pulse from the central source. TORMAC is also capable of generating IR response light curves to observed input optical light curves. [Zach: I think you reminded me a while ago that TORMAC has already done this, need to rephrase] In this paper, we utilize TORMAC to extract additional information about the dust structure of a well-studied AGN, NGC 3783.

NGC 3783 is a prototypical nearby Seyfert 1 galaxy at a redshift of z =? [Zach: cite], implying a luminosity distance of ? Mpc using a standard cosmology of H_0 = 70 km s^{-1} Mpc^{-1}, $\Omega_m = 0.3$, and $\Omega_\Lambda = 0.7$ [Zach: cite]. NGC 3783 has an optical reverberation mapping constrained SMBH mass of M_{BH} =? $\times 10^{?} M_\odot$ [Zach: cite] and an estimated average bolometric luminosity of $L_{bol} = 2.4 \times 10^{44}$? erg s^{-1} [Zach: cite]. NGC 3783 experiences significant optical variability [Zach: feel like I should be able to quantify this], making it an ideal candidate for dust reverberation mapping studies. NGC 3783 has also been extensively studied across the EM spectrum.

IR interferometric observations of NGC 3783 have revealed the presence of a hot, compact, disk-like structure, commonly referred to as the "dusty torus" and a cooler, resolved, extended component in the polar regions (GRAVITY Collaboration 2021). A significant fraction of the IR emission from NGC 3783 comes from this polar dust structure, which we refer to as the "polar bi-cone". Both IR interferometry and SED model fitting **[reffssss]** have been used to gain information about NGC 3783's circumnuclear dust, but this paper serves as the first use of light curves to extract additional information about the circumnuclear dust.

2 METHODS

NGC 3783 has been observed in the B band (optical) as well as the J, H, and K bands (IR response) from 2006 to 2011 with an average cadence of 4.5 days during the observing season (Lira et al. 2011). These resulting light curves have been host galaxy corrected. Since these light curves contain large gaps between observing seasons, the B band light curve is interpolated using the power spectrum derived from the data. The interpolated B band light curve is used as an input to TORMAC (see Section 2.1), representing the variations of the central source. The J, H, and K band light curves are compared to the model light curves TORMAC produces. Therefore, it is used as an input to our MCMC parameter estimation code tormacFit. (see Section 2.2).

2.1 IR Light Curve Model Creation: TORMAC

Since TORMAC has already been introduced in Almeyda et al. (2017), and expanded in Almeyda et al. (2020), here we will provide an overview of TORMAC's primary functions and new capabilities. TORMAC simulates the IR response from the circumnuclear dust surrounding AGN as a 3D ensemble of dust clouds, randomly distributed within user defined geometries. TORMAC models the torus as a clumpy, flared disk with linear scale height. Because classical clumpy torus models are unable to model extended polar mid-IR emission from AGNs, TORMAC is now also capable of simulating clumpy polar dust structures. The number of clouds contained within both structures can be defined by the user, with a canonical value of $N_{cl} = 10^5$ being used in all models presented in this paper. The fraction of the clouds which reside in the torus is given by the user defined parameter $f_{\rm cl}$, where $f_{\rm cl} = 0$ corresponds to all of the clouds residing in the polar dust structure. Both structures are oriented to the observer by the user defined inclination angle i, such that when i

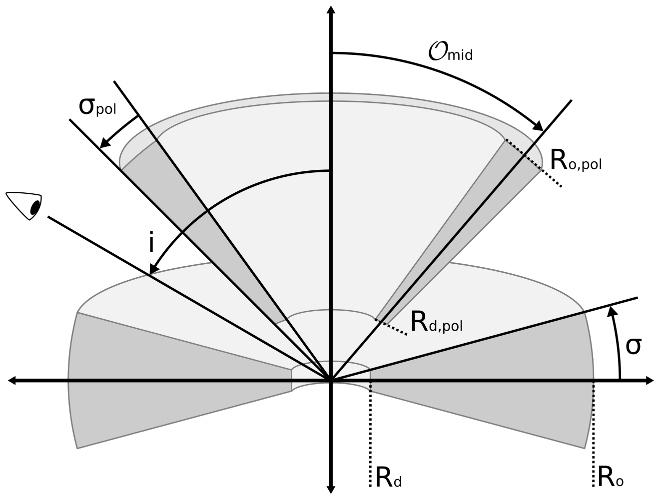

Figure 1.

= 90°, the torus structure is edge on, and when i = 0°, the observer's line of sight passes through the center of the polar dust structure. A cloud's position (in either structure) is defined by its distance from the central illuminating source r, its polar angle θ (where $\theta = 0°$ and $\theta = 180°$ are defined as the polar axes), and its azimuthal angle ϕ.

The torus' geometry extends from the dust sublimation radius R_d to an outer radius R_o. R_o is determined by the user defined parameter Y, where $Y = R_o/R_d$. Radially from the central source, clouds in the torus are arranged following a power-law distribution with user defined index p, where $N_{cl}(r) \propto r^{p-2}$. The half angular width of the torus is determined by the user defined parameter σ. In polar angle, clouds follow a Gaussian distribution centered in the equatorial plane ($\theta = 90°$) and with standard deviation σ, such that the polar angle distribution is defined as:

$$N_{cl}(\theta) \propto \exp[-(\theta - 90°)^2/2\sigma^2]$$

The user also must define the fraction of the torus volume which is filled with clouds, log(Φ). If log(Φ) = 0.01, 1% of the torus' volume will contain clouds.

The polar dust structure is modeled as a hollow bi-cone with the user defined opening angle $O_{\rm mid}$. The polar dust is similarly generated based on the analogous user defined parameters $Y_{\rm pol}$, $p_{\rm pol}$, $\sigma_{\rm pol}$, and log($\Phi_{\rm pol}$). Radially, the polar dust extends from R_d to $R_{o,pol}$, with $Y_{\rm pol} = R_{o,pol}/R_d$. Clouds in the polar dust are also arranged following a power-law distribution with $N_{cl}(r) \propto r^{p_{\rm pol}-2}$. However, in polar angle, the distribution of polar dust clouds is the sum of two Gaussian distributions, centered at $\theta = 0 + O_{mid}$ and $\theta = 180 - O_{mid}$. Both Gaussian distributions have standard deviation $\sigma_{\rm pol}$. This yields the polar angle distribution:

$$N_{cl}(\theta) \propto \exp[-(\theta - O_{mid})^2/2\sigma_{\rm pol}^2] + \exp[-(\theta - (180° - O_{mid}))^2/2\sigma_{\rm pol}^2]$$

The dust clouds are heated either directly by the UV/optical continuum emitted from the accretion disk or, if shadowed [Zach: what is shadowing?], indirectly by the diffuse radiation field produced by the directly illuminated clouds. The illuminating AGN radiation field may be anisotropic, due to "edge darkening" of the accretion disk. Therefore, the AGN luminosity is assumed to have the following polar angle dependence:

$$L(\theta) = [s + (1-s)(1/3)(1 + 2\cos\theta)\cos\theta]\, L_{\rm AGN}$$

(see Netzer 1987), where $L_{\rm AGN}$ is the isotropic bolometric luminosity of the AGN and the "softening parameter" s determines the degree of anisotropy with an s value of 1.0 corresponding to isotropic illumination. In the isotropic case, R_d is a constant value, only dependent on the dust's sublimation temperature $T_{\rm sub}$. However, in the anisotropic case $s < 1$, R_d is instead a function of both the dust sublimation temperature, $T_{\rm sub}$, and θ. For the purposes of calculating R_o, the isotropic R_d is always used in the expression of Y.

[**Insert brief overview of new grain stuff**] [Zach: This needs to contain a reference to sebs paper so we can talk about the cloud grids.] [Zach: Also needs to have the dust sublimation radius equation]

TORMAC also calculates the anisotropic emission from individual clouds, as the illuminated face of a cloud will have a higher temperature than the non-illuminated side, see Section 2.4 of Almeyda et al. (2017). Additionally, the emission from any given cloud is also subject to two "global" opacity effects. The first is cloud shadowing, whereby an outer cloud has its line of sight to the central AGN continuum source blocked by one or more inner clouds, closer to the central source. Clouds that are shadowed in this way are heated indirectly by the diffuse radiation emitted by surrounding directly illuminated clouds (see Almeyda et al. (2017)). Between Almeyda et al. (2020) and this paper, the implementation of cloud shadowing in TORMAC has changed, and is described in detail in Section 2.1.1. The second is cloud occultation, whereby the emitted spectrum of a cloud may be attenuated by clouds intervening along the line of sight to the observer (see Section 2.1 in Almeyda et al. (2020)). [Zach: I definitely think this paragraph should move before the description of illumination, thoughts?]

2.1.1 *Modified Cloud Shadowing*

In Almeyda et al. (2017, 2020), an analytic approximation is used to calculate the probability that a cloud at a given radius will be shadowed. While this approximation may be sufficient for many cases, it often overestimates the number of clouds shadowed, with some extreme cases exacerbating this effect. In the past, TORMAC has used this probabilistic approach to reduce model execution time, but due to general improvements in TORMAC's model execution time, it has become computationally feasible to accurately determine which clouds are shadowed.

We compute each cloud's shadow as a cone that extends from its surface to infinity with a constant solid angle as observed from the central source. To determine if a given cloud is shadowed, we determine which of these cones the cloud intersects, and the area of intersection is calculated and summed. A cloud is considered shadowed if its intersection area is greater than 50% of its total area. If a cloud is considered shadowed, it will only be illuminated by diffuse radiation, whereas if it is not shadowed, it will only be directly illuminated by the central source. [Zach: Make clear that intersection area must be greater than 50% for an individual cone not the sum] For shadowed clouds, the total intersection area is then used to compute the cloud's attenuated radius from the central source according to the equation

$$R_{\rm att} = R_{\rm cl} \cdot \exp\left(A_{\rm int}/2A_{\rm cl}\right)$$

where $R_{\rm cl}$ is the physical radius from the central source of the cloud, $A_{\rm int}$ is the total intersection area, and $A_{\rm cl}$ is the area of the cloud. The attenuated radius is then used to determine the amount of diffuse radiation expected, based off of a precomputed cloud grid. [Zach: What is this precomputed grid? Say that it is based off of the clouds distance and wavelength] [Zach: need to better explain what the "attenuated radius" means]

2.2 MCMC Model Fitting Using tormacFit

[Zach: Would probably be better to say something along the lines of: To rapidly sample the goodness of fit of our models across the parameter space to the ground truth observed data, we use mcmc...]As a way to estimate the constraints on the parameters for which we generate TORMAC models, we employ Markov Chain Monte Carlo (MCMC) modeling to fit the IR response models generated by TORMAC to the observations. We use a code entitled tormacFit, which is a heavily modified version of clumpyDREAM (Sales et al. 2015), to perform the MCMC search. [Zach: Probably should just cite pydream] The fitter calculates the likelihood using a χ^2 test using all of the points in the observed light curve. [Zach: Can revise this to better show mathematically what our log liklihood looks like] Since TORMAC is capable of calculating light curves for multiple wavelengths, and multiple observed wavebands (J, H, and K) are available for NGC 3783, tormacFit was made to handle multi-wavelength fitting. The likelihoods calculated for each wavelength are multiplied to arrive at the final likelihood used for the MCMC search. [Zach: Can just say that the code jointly fits the wavelengths] For all of the posterior distributions presented in Section 3, we fit to both the H and K bands. We were unable to properly fit the light curves to the J band, explained in Section 4. [Zach: This is a poor location to put this info...]

We introduce two additional parameters to our MCMC search: log(amp) and Sys. Unc. (%). The log(amp) parameter is a monochromatic scaling factor applied to the model lightcurve, used to . log(amp) values far from unity represent limitations of TORMAC to reproduce the true observed flux, or additional sources of extinction which have not been considered. The Sys. Unc. (%) parameter is systematic uncertainty added to the uncertainty on the fluxes for the observed light curve, and represents additional random, yet unexplained fluctuations introduced during reprocessing of the optical radiation. [Zach: this is not a good way to explain systematic uncertainty]

To perform the MCMC search, we calculate a set of TORMAC models which sample the parameter space on a regular grid. For MCMC proposal evaluations which do not lie on one of these grid points, the model light curve is nd-linearly interpolated between adjacent grid points. Therefore, we impose that all of our model grids are "regular", meaning that the set of models we run is the cartesian product of all of the sets containing a respective parameter's values to be tested, i.e., there are no gaps in the grid.

3 RESULTS

As discussed in Section 2.1, TORMAC is able to model the IR response from an AGN, producing the response IR light curve from the input optical light curve. In this section we focus on the resulting parameter estimations from a variety of modeled dust cloud distributions to the observed light curves. We begin by evaluating a regular grid [Zach: I define "regular grid" only below] of models which only vary the parameters in the torus dust distribution (Y, σ, p, $\log(\Phi)$, as well as i). Because we know that NGC 3783 likely has a polar dust distribution, we also construct a model grid which jointly simulates a torus and polar dust distribution, again varying Y, σ, p, $\log(\Phi)$, & i, but also varying $Y_{\rm pol}$, $\sigma_{\rm pol}$, $p_{\rm pol}$, $\log(\Phi_{\rm pol})$, and $O_{\rm mid}$. For each of our model grids, we evaluate both isotropic ($s = 1.0$) and anisotropic ($s = 0.1$) illumination cases. In all cases, we generate TORMAC simulated response light curves in both the H and K bands.

After evaluating the model grid, we use Dynesty's dynamic nested-sampling algorithm, which draws parameters from a uniform prior

Table 1. Parameter values used to construct the model grids in this work. Parameter values represent points on a regular grid with no gaps, ie. all possible combinations of parameters are evaluated.

Parameter	Torus Only	Torus & Polar
Y	10, 25, 50, 75, 100, 500	2, 5, 10, 25, 50
σ	5, 10, 30, 45	2, 5, 10, 20
p	-2, -1, 0, 1, 2	-1, 0, 1, 2
$\log(\Phi)$	-1, -2, -3, -4	-1, -2, -3, -4
i	0, 30, 45, 60, 90	0, 30, 45, 60, 75
s	1.0 (iso), 0.1 (aniso)	1.0, 0.1
$Y_{\rm pol}$	–	5, 50, 100
$\sigma_{\rm pol}$	–	2, 10, 20
$p_{\rm pol}$	–	-1, 0, 1, 2
$\log(\Phi_{\rm pol})$	–	-1, -2, -3, -4
$O_{\rm mid}$	–	10, 25, 40
$f_{\rm cl}$	–	0.333, 0.667
numclouds	50000	100000

and computes their corresponding likelihood values. We need to sample light curves continuously within the maximum and minimum bounds of each parameter our grid (the prior space). To do this, we linearly interpolate between adjacent TORMAC models in our grid. This is an approximation which we assess in Section 4. To calculate the likelihood, we first modify the interpolated response light curves by multiplicatively scaling both light curves by a single scaling factor which maximizes the sample's joint gaussian likelihood to both the H and K band IR observed response light curves. We then jointly calculate the gaussian likelihood to the observed H and K band IR response light curves of NGC3783. Across all grids and cases presented in this work, we use the same configuration of Dynesty dynamic nested sampling algorithm. We use a Dynesty configuration with 2000 live points and a target of 600000 posterior samples as our stopping criterion.

3.1 Torus Only Models

AGN circumnuclear dust has typically been modeled as a single compact disk component [Zach: cite?]. We investigate if a single torus component is sufficient to model the observed IR light curves of NGC 3783 [Zach: last two sentences actually go in intro?]. The geometric parameters of the torus (Y, σ, i, and s) as well as the structural parameters (p and $\log(\Phi)$) were varied independently, such that all possible combinations of the parameter values are sampled, herein referred to as a "regular grid". The model grid is finely sampled over these parameters (relative to the torus + polar grid), with a complete set of parameter values enumerated in Table 2.

For both the anisotropic and isotropic illumination cases, we separately performed dynamic nested sampling using Dynesty. This was done to distinguish the influence from the type of illumination of the dust distribution from the rest of the parameters. We find that the anisotropically illuminated torus only model grid better describes the observed H and K band light curves of NGC3783, with a reduced statistic of $\chi^2 = 5.14$ and $\chi^2 = 3.93$ for the isotropic and anisotropic grids respectively. However, both of these model grids are a generally poor description of the data. The posterior distribution estimated from the anisotropic model grid are shown in Figure 2. For the posterior distributions presented in this work, we choose to visualize both the 1-dimensional (along the diagonal) and 2-dimensional (in the lower triangle) posterior distributions. The 1-dimensional posterior distributions represent the posterior of each parameter varied in the torus only model grid ($\log(\Phi)$, p, i, Y, σ), with the value listed

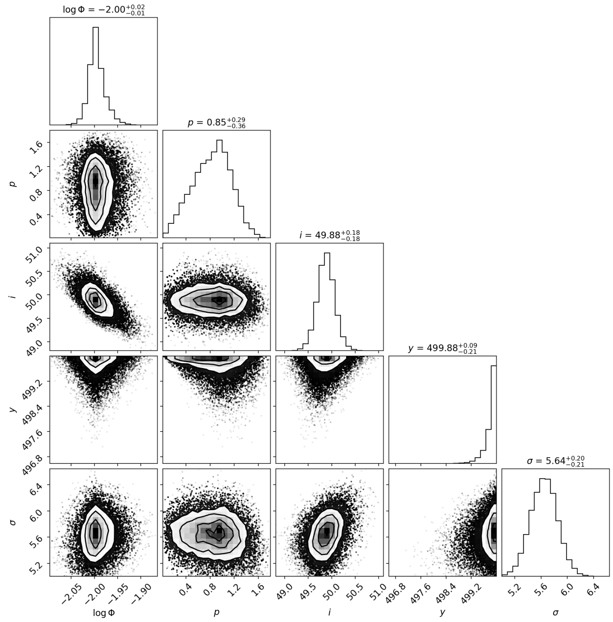

Figure 2. Estimated posterior distributions of directly fitted parameters from the torus only model grid with $s = 0.1$. 1-dimensional posterior distributions for the parameters $\log(\Phi)$, p, $\imath$, Y, and σ lie on the diagonal, with the indicated value above each distribution corresponding to the maximum of the posterior distribution in that dimension. 2-dimensional posterior distributions for all combinations of parameters are shown in the lower triangle. The contours enclose the 0.5, 1, and 2 σ regions of the posterior distribution. To avoid over crowding of points in the 2-dimensional posteriors, the shade of the region enclosed by the 2 σ contour indicates the relative density of posterior samples.

above each panel corresponding to the maximum of the posterior distribution for that given parameter. The 2-dimensional posterior distributions show the relationship between combinations of parameters. When a parameter or combination of parameters is well constrained, the posterior distribution for that parameter or combination of parameters will be roughly that of a gaussian, such as is the case for i, σ, and the combination of i and σ in the anisotropic illumination case of the torus only model grid. i and $\log(\Phi)$ are related, indicated by the skew present in the 2-dimensional posterior. This generally indicates that models with a closer to face on inclination are better supported when a greater fraction of the torus' volume contains clouds than for higher inclinations. Y is not constrained in the torus only model grid with anisotropic illumination, indicated by the peak posterior value being very close to the edge of the posterior volume ($Y = 500$), and the posterior distribution not being close to zero at the edge of the posterior volume. With the notable exception of Y, all other parameters are constrained. p is relatively weakly constrained, indicating that models with more centrally concentrated clouds and models with clouds concentrated at larger radii are able to describe the infrared response of NGC3783 nearly equally as well. For this set of models, NGC3783's infared response is best modeled by a moderately inclined, angularly thin, radially thick torus with roughly 1% of the torus' volume containing clouds.

3.1.1 Measured Variables

In addition to the tunable parameters of the TORMAC model which have been directly fit by dynamic nested sampling, we also obtain

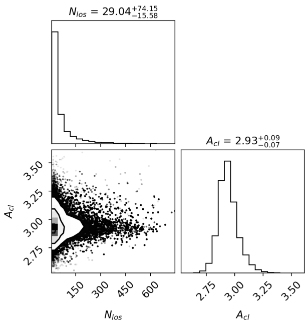

Figure 3. Same as Figure 2, but for the measured parameters $N_{\rm los}$ and $A_{\rm cl}$ not directly fit through dynamic nested sampling.

measurements of other physically relevant quantities which are fixed for a given set of directly fit parameters. $N_{\rm los}$ is defined as the number of clouds on average intersected by rays originating at the central source and which radially extend to infinity and which pass through the torus distribution, and is directly proportional to the optical depth of the distribution. $A_{\rm cl}$ is defined as the ratio of the surface areas of a single cloud relative to the inner surface area of the torus, with larger values indicating that the simulated response is more likely to be dominated by the fewer, but larger directly illuminated clouds. For each of the posterior samples in Figure 2, we calculate $N_{\rm los}$ and $A_{\rm cl}$. For the anisotropic illumination case of the torus only model grid, we represent the samples from the estimated posterior distributions of $N_{\rm los}$ and $A_{\rm cl}$ as a corner plot in Figure 3. The peak of posterior values for $N_{\rm los}$ and $A_{\rm cl}$ are very large, meaning that this torus distribution is optically thick and has very few clouds which are directly illuminated.

3.2 Variable Torus and Polar Dust Models

While we initially chose to limit our model to only simulate the dust response from a single torus distribution, many AGN, including NGC 3783, have been shown through interferometry to also have extended warm dust IR emission perpendicular to the torus. Thus, we chose to include a simulated polar dust component in addition to a simulated torus component. This substantially increases the complexity of our model grid, resulting in roughly double the number of free parameters. Therefore, we choose to generate fewer TORMAC models at extreme (and not generally not physically supported) parameter values. First, we limited the inclination to be $i < 75°$, since for $i > 75°$ NGC 3783 would not have been able to be reverberation mapped. However, we will see that this restriction on i does not forbid the introduction of dust at very high inclinations. Additionally, we chose to not explore models beyond $Y_{\rm pol} = 100$, $Y = 50$ due to the unphysically large clouds that the $Y > 50$ models produce and that these models have never been favored in the previous posteriors. To ensure that we included small as well as extended tori in this model set, we also chose to decrease the minimum Y value examined from 10 to 2. To limit numerical noise from having discrete clouds in our model, we chose to double the number of simulated clouds in the TORMAC models from 50000 to 100000, in order to keep the number of clouds in each distribution of order equal to the torus only case. To examine the effect that the number of clouds in each distribution has on the resulting light curve and posterior distribution, we, in addition to still evaluating both the isotropic and anisotropic illumination cases, now also evaluate separate grids with 1/3 of the simulated clouds being placed in the torus ($f_{\rm cl} = 0.333$) and 2/3 of the simulated clouds being placed in the torus ($f_{\rm cl} = 0.667$). This results in a total of four cases. As with the torus only grid, all parameter values are evaluated on a regular grid with grid points enumerated in Table 2.

Of the four cases we evaluate, we choose to focus our analysis on the anisotropically illuminated case with 1/3 of the clouds being placed in the torus dust distribution, as this case is the best description [Zach: I stopped writing mid sentence here]. The posterior distribution of the directly fit parameters for the isotropic case with 1/3 of the clouds in the torus are shown in Figure 4. Like the torus only case, we find that all of the tested torus parameters except for Y are constrained, with the parameter values weighted most heavily in the posterior distribution being relatively invariant to the addition of the polar dust component, ie. a moderately inclined, angularly thin, radially extended torus with roughly 1% of the torus' volume containing clouds. Notable exceptions include the significantly stronger constraint on p and σ, as well as a shift towards lower inclinations. The shift towards lower inclinations could be explained by the addition of the–highly inclined–polar dust distribution. Y is once again favors the largest possible value, in this grid, 50. Many of the parameters for the polar dust distribution are also constrained, with the exception of $Y_{\rm pol}$ and $\sigma_{\rm pol}$, which both are weighted heavily towards their upper bounds: 100 and 20°. In relation to the torus distribution, the polar dust distribution's $\log(\Phi_{\rm pol})$ is significantly higher than $\log(\Phi)$, with the posterior distribution heavily weighted close to, but still constrained within the upper bound of 10% of the polar dust distribution's volume containing clouds. Additionally, the polar dust component is significantly angularly extended relative to the torus. However, on the other strutural parameters (p & $p_{\rm pol}$, Y & $Y_{\rm pol}$) the highest posterior weights are roughly analagous between the torus and polar dust. NGC3783's IR response is best modeled with $O_{\rm mid} \approx 30°$ and $i = 30°$, which, when paired, results in dust clouds being placed along the line of sight of the observer.

3.2.1 *Measured Variables*

As with the torus only model grid, we also calculate $N_{\rm los}$ and $A_{\rm cl}$ for each of the posterior samples, but in addition, we are able to also calculate $N_{\rm los,pol}$ and $A_{\rm cl,pol}$, which are the number of clouds along the line of sight and the cloud area ratio for the polar dust distribution, respectively. We also determine the anglular intersection between the torus and polar dust distributions, which we refer to as $\theta_{\rm int}$, with values > 0 indicating angular intersection between the distributions. Finally, we also calculate the ratio of the cloud size for the polar dust distribution to the cloud size for the torus distribution, $R_{\rm c,pol}/R_{\rm c}$.

For the anisotropic illumination case with 1/3 of the clouds in the torus, we show the estimated posterior distribution of $N_{\rm los}$, $A_{\rm cl}$, $N_{\rm los,pol}$, $A_{\rm cl,pol}$, $\theta_{\rm int}$, and $R_{\rm c,pol}/R_{\rm c}$ in Figure 5. The peak of posterior values for $N_{\rm los}$ and $A_{\rm cl}$ are both significantly lower than for the torus only case, meaning that this torus distribution is optically thinner and has more clouds which are directly illuminated. The polar dust distribution is also best supported with a significantly lower $N_{\rm los}$ value than would be indicated by the torus only model grid. However, the peak of posterior values for $A_{\rm cl,pol}$ are significantly larger than $A_{\rm cl}$, meaning that this polar dust distribution is also optically thinner than the torus in the torus only model grid, but also suffers from having very few directly illuminated clouds. We also find that the combinations of $N_{\rm los}$ and $A_{\rm cl}$ as well as $N_{\rm los,pol}$ and $A_{\rm cl,pol}$ are both positively correlated, which makes sense as they are fundamentally related quantities. Interestingly, we also find that $R_{\rm c,pol}/R_{\rm c}$ and $N_{\rm los}$ are negatively correlated, which implies that reasonable prescriptions to the observed IR response of NGC3783 can be achieved over a range of cloud area ratios. We also find that the preferred cloud area ratio

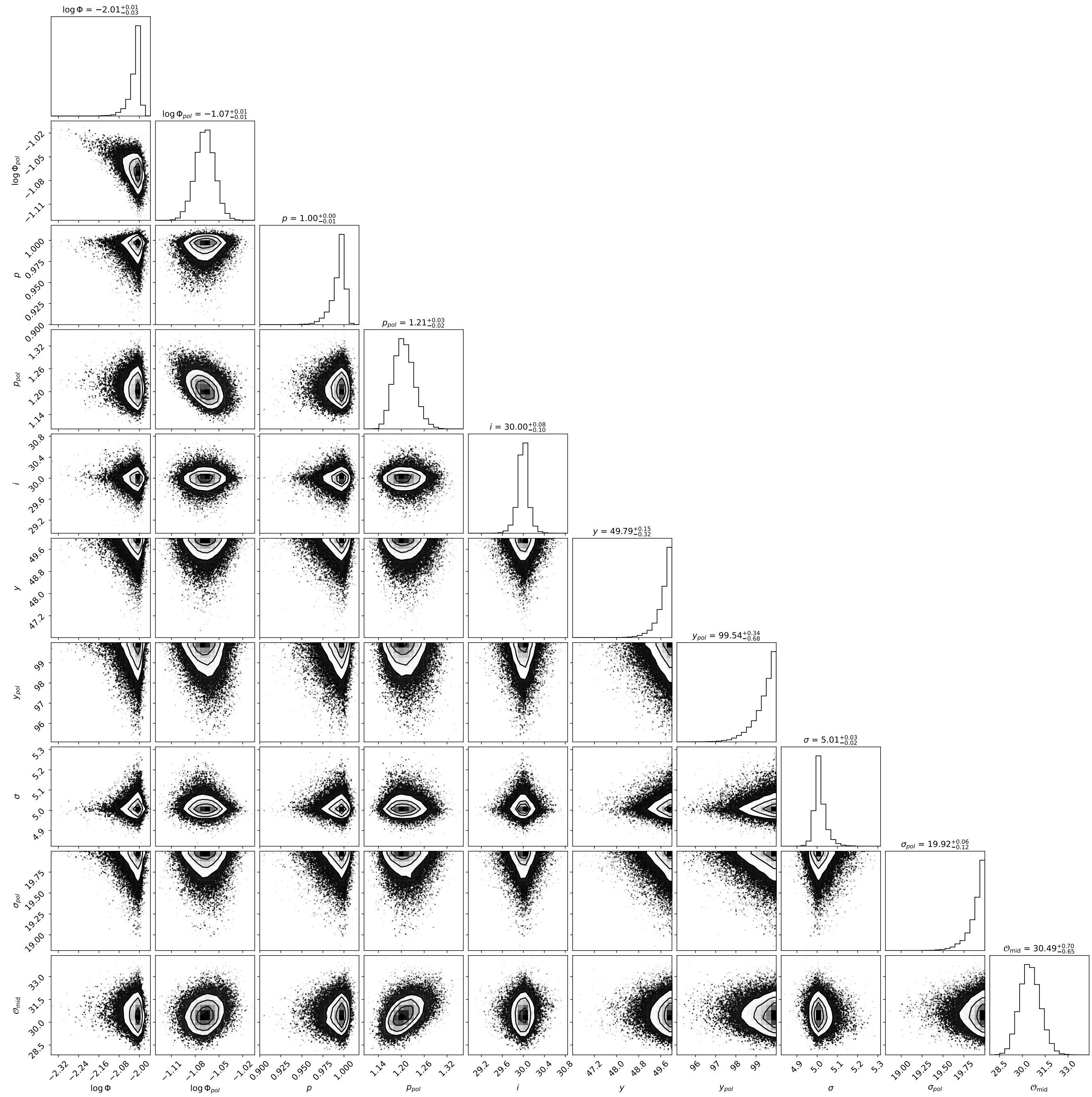

Figure 4. Analog of Figure 2, but for the torus & polar dust model set for $s = 0.1$ and $f_{\rm cl} = 0.333$. Again, the 1-dimensional posterior distributions lie on the diagonal, the indicated value above each distribution corresponds the maximum of the posterior distribution in that dimension, and the 2-dimensional posterior distributions for all combinations of parameters are shown in the lower triangle.

is > 1, indicating that the polar dust clouds are larger than the torus clouds.

3.2.2 *Limitations on the Prior Space*

We also used Dynesty to perform dynamic nested sampling on various subsets of the overall prior space. We do this to better understand how our estimated maximum likelihood model light curve and associated posterior distributions are affected by choice of several observational constraints. In Figure 6 and Figure 7, we show the estimated posterior distributions for the directly fit parameters and the measured parameters, respectively, while applying all of the following prior space restrictions simultaneously: $1 < N_{\rm los} < 15$, $1 < N_{\rm los,pol} < 15$, $A_{\rm cl} < 0.01 \equiv 1\%$ of torus illuminated surface area, $A_{\rm cl,pol} < 0.01 \equiv 1\%$ of polar dust illuminated surface area, $\theta_{\rm int} < 0$, and $R_{\rm c,pol}/R_{\rm c} < 1$. [Zach: understand that will probably need to justify these choices, but not informed on why all of these

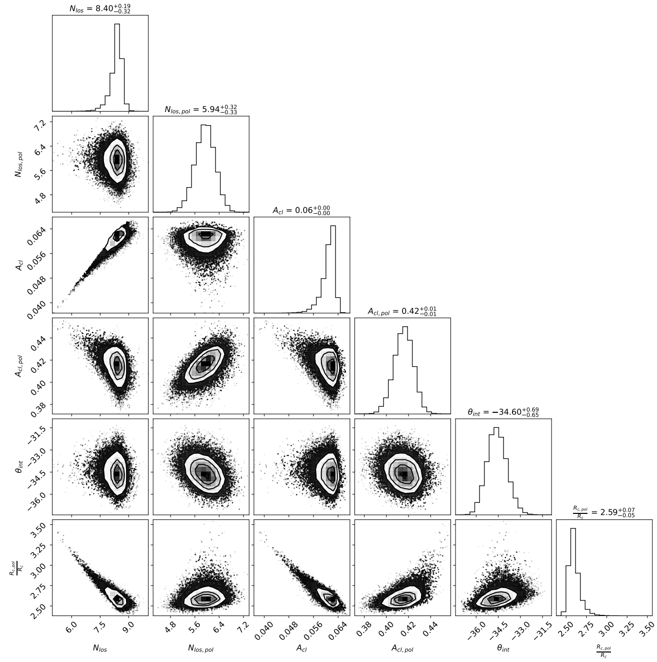

Figure 5. Analog of Figure 3, but for the torus & polar dust model set for $s = 0.1$ and $f_{\rm cl} = 0.333$ with directly fitted parameters shown in Figure 4. The 1-dimensional posterior distributions for the measured parameters $N_{\rm los}$, $A_{\rm cl}$, $N_{\rm los,pol}$, $A_{\rm cl,pol}$, $\theta_{\rm int}$, and $R_{\rm c,pol}/R_{\rm c}$ lie on the diagonal, with the indicated value above each distribution corresponding to the maximum of the posterior distribution in that dimension. The 2-dimensional posterior distributions for all combinations of parameters are shown in the lower triangle.

specific numeric values are chosen.] We find that the application of these prior restrictions significantly limits the ability of TORMAC to reproduce the observed IR response of NGC3783, with the maximum likelihood model having a reduced statistic of $\chi^2 = 4.18$. Additionally, we find that the application of these prior restrictions significantly changes the estimated posterior distributions for both the directly fit parameters and the measured parameters. For the directly fit parameters, we find that the application of these prior restrictions results in a shift towards higher (but still constrained) inclinations, and lower (but now constrained) polar angular width. However, many of the previously constrained parameters are now unconstrained: $\log(\Phi)$ & $\log(\Phi_{\rm pol})$ both hit their upper bound of 10% volume filled in their respective distributions, p and $p_{\rm pol}$ are now unconstrained towards clouds being placed at the outer edge of their respective dust distribution, and Y and $Y_{\rm pol}$ are now unconstrained at their lower bound, and $O_{\rm mid}$ is unconstrained at its lower bound. This results in a dense (in $\log(\Phi)$), compact, highly inclined torus and polar dust distribution with small angular thickness. [Zach: note for discussion: this has gotta be because to meet our prior restrictions, now Y and $Y_{\rm pol}$ must be unnaturally small: the response is truncated at long time delays = worse fit. soln?: make variable size clouds so we can satisfy large vff but without having absolute BRICKS at the inner edge.] We find that σ and $\sigma_{\rm pol}$ are related, with models having smaller angular widths for the torus region and polar region, while maintaining ($\sigma/\sigma_{\rm pol} \approx 0.5$), represented in the posterior distribution. This relationship is also present in the posterior distribution of the indirectly fitted parameters, shown in Figure 7 as a tail present in nearly all of the parameters. Interestingly, despite previous posterior distributions favoring values of $N_{\rm los}$, $N_{\rm los,pol}$, $A_{\rm cl}$, and $A_{\rm cl,pol}$ which largely exceeded the prior restrictions, all of these parameters do not favor their upper bound, but instead are weighted towards the middle of the allowed prior space. This suggests that while the application of these prior restrictions does significantly change the estimated posterior distribution, it does not simply shift the previously estimated posterior distribution to be truncated at the upper bound of the prior space.

4 DISCUSSION

4.1 Comparison to Other Models of NGC 3783

Parameter estimates from this technique tend to correlate with predictions made by other methods. IR interferometric observations predict that the torus component of the dust has an inclination that is close to face-on to the observer ($i < 40°$) and a radial extent of $\tilde{0}.07$pc for the hot inner edge of the dust (GRAVITY Collaboration 2021). Kinematic simulations of NGC 3783 find a best fit to a polar bi-cone roughly perpendicular to the torus ($i_{\rm pol} \approx 60$-$75°$ with $i_{\rm tor} \approx 35$-$40°$, an offset of $\tilde{6}5$-$85°$) with $\sigma_{\rm pol} \approx 7$-$10°$, and a half opening angle of $\tilde{3}0$-$50°$ (Fischer et al. 2013)[Zach: ; Müller-Sánchez et al. 2011]. Our models tend to suggest that higher inclination solutions may be allowable, but otherwise our findings are consistent with their results. Dust continuum SED models presented in Hönig and Kishimoto (2017) favor moderate inclinations between 15° and 60°, p values between –1 and 1, and $p_{\rm pol} > 1$.

In the previous section, we examined the posterior distributions for the structure and geometry of a two component (torus and polar bi-cone) dust distribution model. We found that by increasing the complexity of the model grids used to perform the MCMC search, we were able to better model the IR response of NGC 3783.

4.2 Parameter Estimation of NGC 3783

We find that not all tested parameters have equal impact on the best fit IR response light curve, and are thus more difficult to constrain using this technique. Of the parameters that do most strongly impact the response light curve, we have yet to constrain only some. [Zach: General statement not super useful here. This paragraph almost feels like it should be at the end of this subsection]

We consistently find the strongest constraints for p ($\tilde{0}$) and $p_{\rm pol}$ ($\tilde{0}$), the radial cloud distribution power law indices. Lower p values tend to result in light curves that respond on too short a time delay resulting in a light curve that is too "sharp". This is likely due to most of the responding clouds being very close to the dust sublimation radius. Conversely, higher p values tend to overly smooth the resulting light curve, with the same justification as for low p values.

The inclination (i) of the distribution is also generally well constrained to moderate inclinations (30°-60°), with the caveat that completely edge-on models ($i = 90°$) were excluded as priors on the final set of models as they are generally unphysical to reverberation map. Performing reverberation mapping relies on the observer's view of the central source being optically thin to the dust. In edge-on models, the central source would be completely obscured by the torus. These edge-on models, under certain conditions, resulted in degeneracies with models with other less extreme inclinations, motivating their exclusion. We find that inclination has more complex relationships with the other parameters, particularly the opening angle $O_{\rm mid}$. This complexity can likely be attributed to discontinuities between optically thin and optically thick lines of sight to the inner regions of the dust, where the response is strongest.

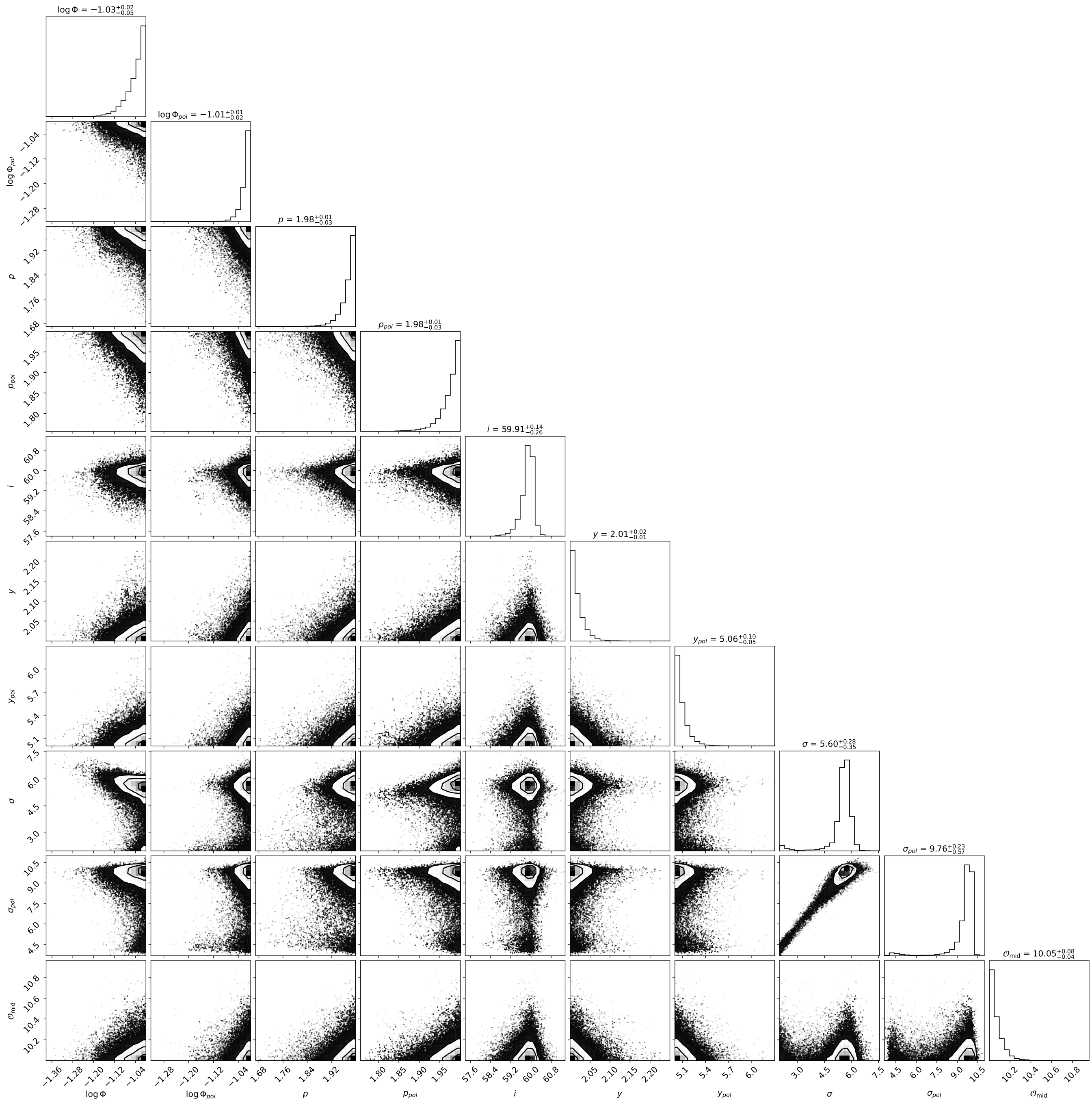

Figure 6. Direct analog of Figure 4, but with additional prior space restrictions of $1 < N_{\rm los} < 15$, $1 < N_{\rm los,pol} < 15$, $A_{\rm cl} < 0.01 \equiv 1\%$ of torus illuminated surface area, $A_{\rm cl,pol} < 0.01 \equiv 1\%$ of polar dust illuminated surface area, $\theta_{\rm int} < 0$, and $R_{\rm c,pol}/R_{\rm c} < 1$.

We find that model grids tend to favor smaller Y values. Hot dust at approximately 1000-2000K responds most strongly in the near-IR H and K bands. Only dust close to the dust sublimation radius ($\lesssim 2R_d$) can reach these temperatures on their illuminated side, suggesting that the best fit Y and $Y_{\rm pol}$ values we find may be more closely related to controlling the overall volume of the dust distribution and thus the cloud sizes. This also suggests that it may be possible to use this technique to further constrain the dust distribution at larger radii using light curves at longer wavelengths. We find that across model grids, σ and $\sigma_{\rm pol}$ have a proportionally smaller impact on the response light curve and have not been constrained by our MCMC search. Comparing light curves where σ and $\sigma_{\rm pol}$ are varied, the shorter timescale features tend to be muted as σ and $\sigma_{\rm pol}$ increase. However, in further investigations, we would expect the angular width to also relate to the inclination in a similar way to $\mathcal{O}_{\rm mid}$. [Zach: General ideas for this section: Could definitely figure out parameter importances!]

- Previous works generally support the conclusions drawn from "Torus 1" of the variable wind model set.

Table 2. Comparison of parameter estimates from different studies of NGC 3783. The first five columns show estimates from previous studies using different methods, while the last two columns show the estimates from this work using TORMAC with a torus only model and a torus plus polar bi-cone model.

	Hönig & Kishimoto (2017)	GRAVITY Collaboration (2021)	Lyu & Rieke (2018)	Bentz et al. (2021)	Fischer et al. (2013)	This Work (Torus Only)	This Work (Torus + Polar)
Y	N/A		500				
σ	<17°	50-13		40-15			
p	-1-1		0-1				
$\log(\Phi)$							
i		39-13°	30-12°		35°		
$Y_{\rm pol}$							
$\sigma_{\rm pol}$				10°			
$p_{\rm pol}$	>1						
$\log(\Phi_{\rm pol})$							
$O_{\rm mid}$	Inconclusive			50			

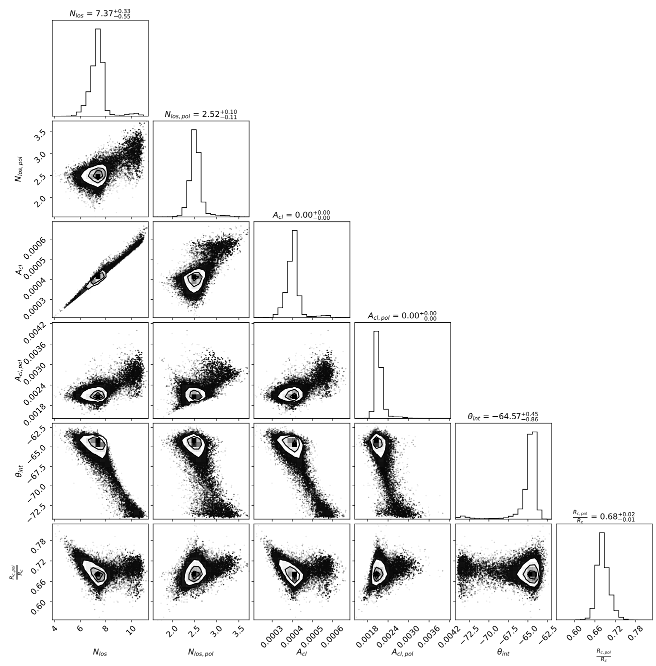

Figure 7. Analog of Figure 5, but with additional prior space restrictions of $1 < N_{\rm los} < 15$, $1 < N_{\rm los,pol} < 15$, $A_{\rm cl} < 0.01 \equiv 1\%$ of torus illuminated surface area, $A_{\rm cl,pol} < 0.01 \equiv 1\%$ of polar dust illuminated surface area, $\theta_{\rm int} < 0$, and $R_{\rm c,pol}/R_{\rm c} < 1$.

- It seems that grids must be relatively finely sampled for the influence from a non-delta transfer function to be determined.
- The parameter space should only contain feasible models, as non-physical / generally unsupported models can cause degeneracy with feasible models.

4.3 Limitations & Future Work

4.3.1 *Saturation Effects*

We now discuss multiple effects which we classify as "saturation effects" as they all have similar effect on the resulting simulated response, namely, at high relative luminosities they each result in a stunting of the response (saturation) which results in significant deviation from the observed data.

One such effect which is well understood is when clouds become a sufficiently large such that only a small number of clouds ($\lesssim$ 100) are able to react directly. This is typically caused by large volume filling factors, large y values, and exacerbated by negative p values. These models are nonphysical due to the clouds' large size particularly at close to the inner edge, but the parameters used may not be non-physical. This could be addressed by introducing a new parameter to describe the radial dependence of cloud volume, or by significantly increasing the number of clouds in the simulation.

DESCRIBE THE DEVIATION IN FIGURE 5 It is hypothesized that this deviation is caused by the fact that TORMAC currently simulates dust sublimation as a function of temperature only, when in reality it is also a time dependent process. In reality, as clouds at the inner edge of the dust distribution are pushed inside of the expanding dust sublimation radius, the inner edge of the clouds (which receive the majority of the heating from the central source) begin to sublimate away, while the interior of the cloud raises in temperature. Instead, since TORMAC simply caps the front edge of the cloud at the dust sublimation temperature, the interior of the cloud is capped at a lower temperature than would be expected, resulting in a saturation of the clouds during periods of high emission from the disk.

- Produce more models with similar parameters to those explored in the third set of models, and iteratively refining the results by adding more in-between models.
- Model dust sublimation as a function of luminosity and time, may influence the response at high luminosities
- Model the polar bi-cone as parabolic and or hyperbolic
- Introduce variable cloud size as a function of radius

5 CONCLUSION

- Torus and polar bi-cone models seem more strongly supported, agrees with observations.
- These models tend to agree with and extend previous attempts to model NGC 3783.
- IR response is very sensitive to some parameters (p, y, vff) but not to others (sig, i).
- Because some parameters have opposite effects on the IR response, it can be difficult to tell some models apart.

ACKNOWLEDGEMENTS

DATA AVAILABILITY

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APPENDIX A: SOME EXTRA MATERIAL